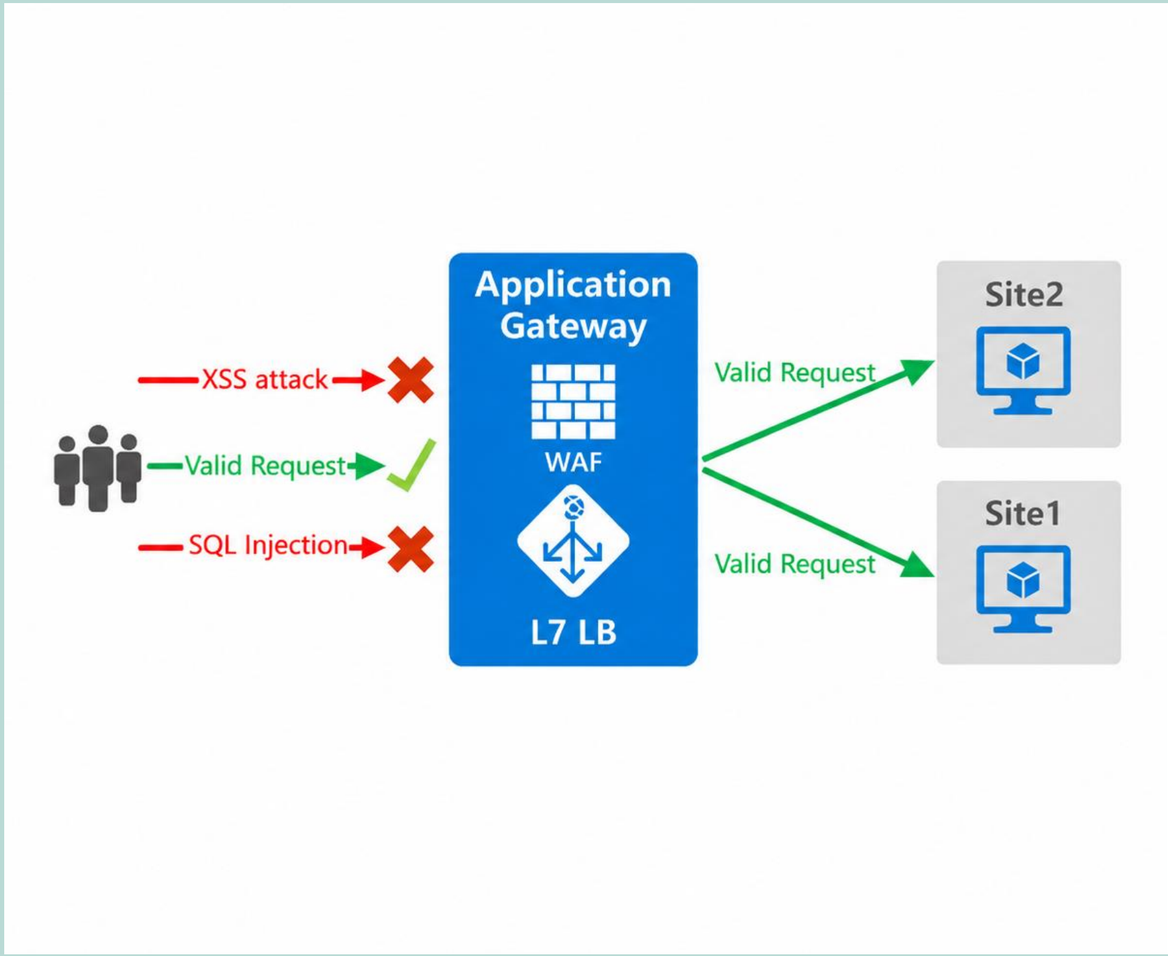
AZURE APPLICATION GATEWAY with Web Application Firewall

Projetct Statement:

“Securing Web Applications with Azure Application Gateway”

Web applications can face many security problems and unwanted traffic. This project uses Azure Application Gateway with Web Application Firewall (WAF) to protect the application. It checks incoming requests, blocks harmful traffic, sends users to the correct backend server based on the URL, handles HTTPS securely, and checks if the backend servers are working properly.

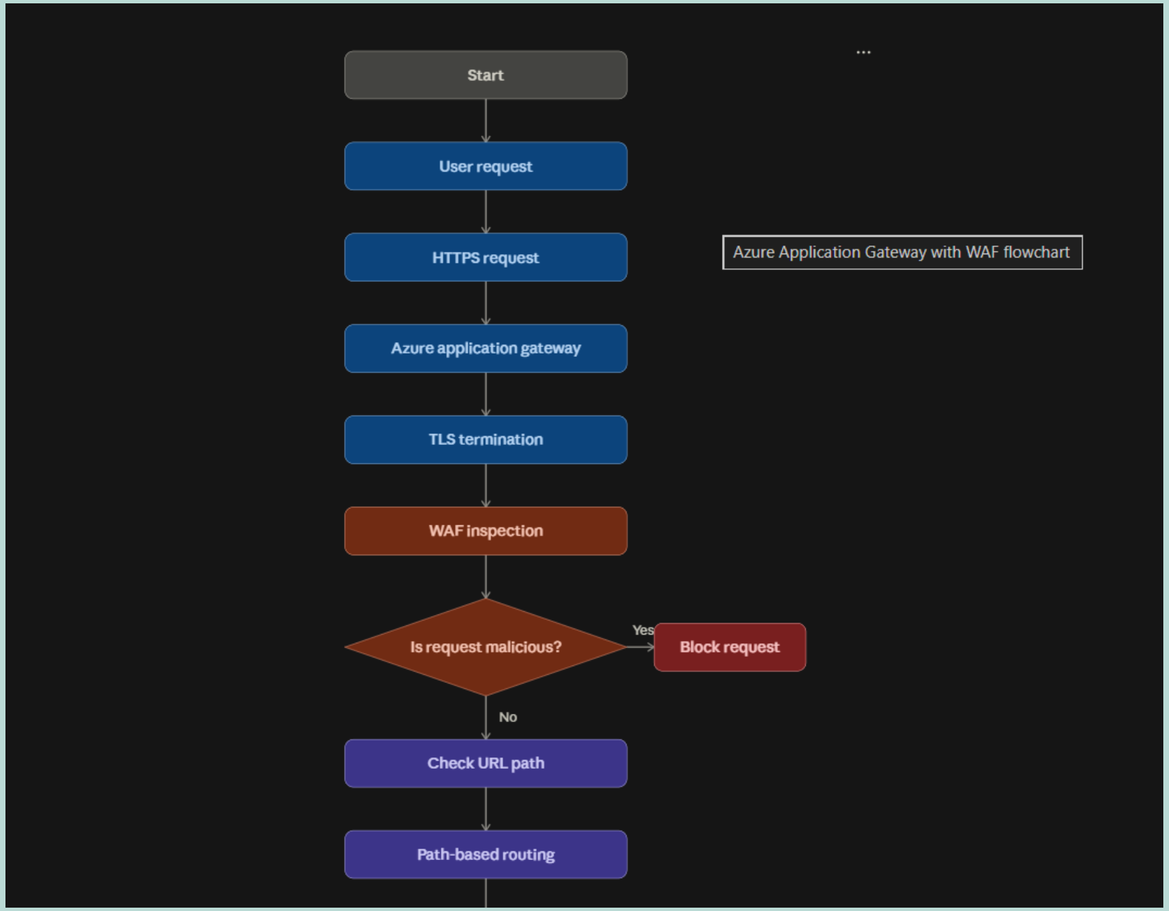


PROJECT OBJECTIVES

The main objectives of this project are:

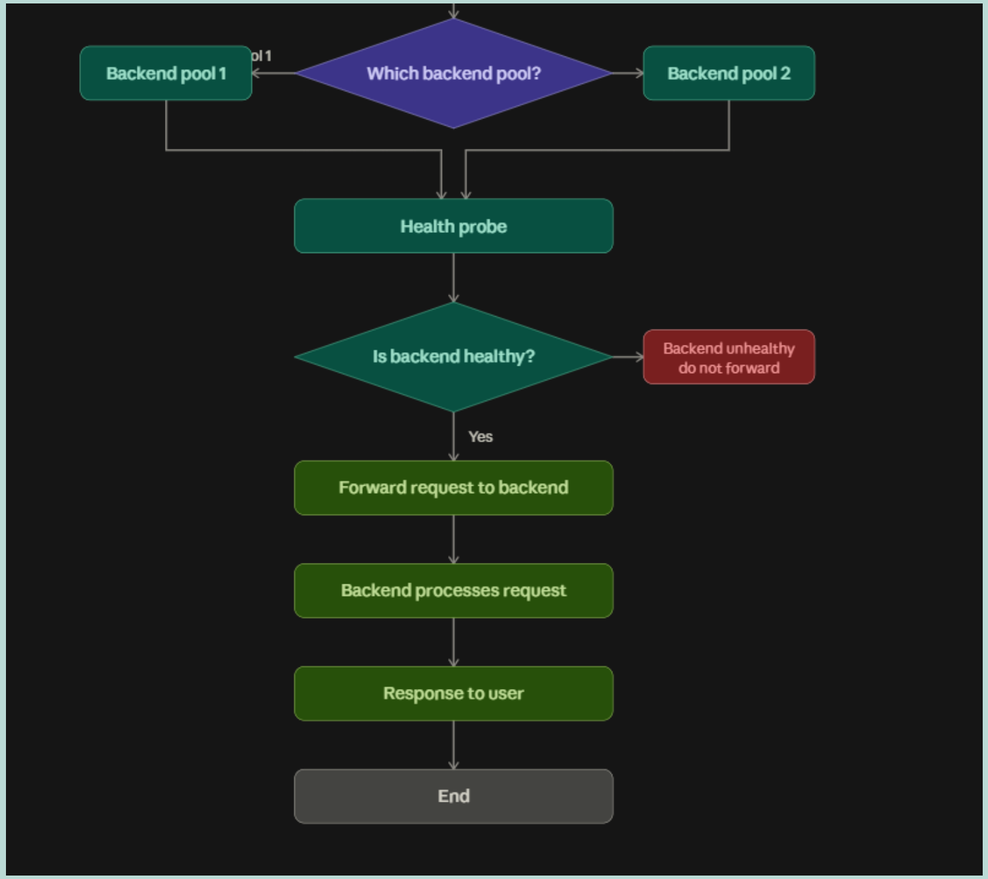
- Protect web applications from common attacks.
- Filter incoming traffic using Web Application Firewall.
- Route requests to the correct backend pool based on the URL.
- Secure HTTPS traffic using TLS termination.
- Monitor backend servers using health probes.
- Detect suspicious requests and block them in Prevention mode.

PROJECT WORKING FLOW



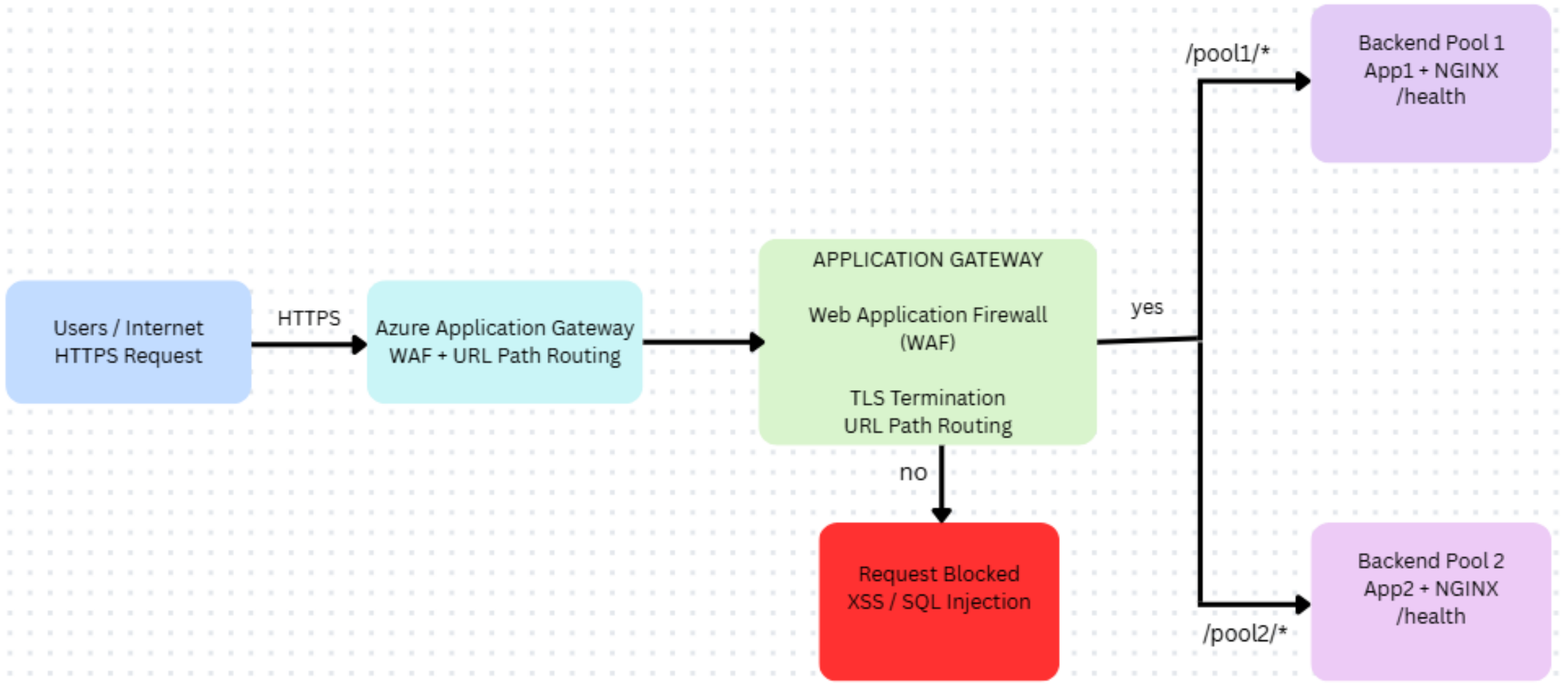
The user first sends an HTTPS request to the Azure Application Gateway. The gateway performs TLS termination and passes the request to the Web Application Firewall (WAF) for security inspection. The WAF checks the request against its security rules. If the request is identified as malicious, it is blocked. If the request is valid, the gateway checks the URL path and uses path-based routing to select the appropriate backend pool.

PROJECT WORKING FLOW



After the WAF allows a valid request, the Application Gateway checks the URL path and selects the appropriate backend pool. The request is sent to either Backend Pool 1 or Backend Pool 2. The gateway then checks the backend health. If the selected backend is healthy and returns a valid response, the request is forwarded and processed. The backend response is then returned to the user. If the backend is unhealthy, the gateway does not forward traffic to that unhealthy backend

Project Architecture



PROPOSED SOLUTION

Our solution uses Azure Application Gateway with Web Application Firewall (WAF) as a secure entry point for the web application.

- All user requests first pass through the Application Gateway.
- WAF checks the requests and blocks harmful traffic.
- HTTPS traffic is securely handled using TLS termination.
- Requests are routed to the correct backend pool based on the URL path.
- Health probes check whether the backend applications are healthy.
- Valid requests are forwarded to the appropriate backend application, while suspicious requests can be detected or blocked by the WAF.

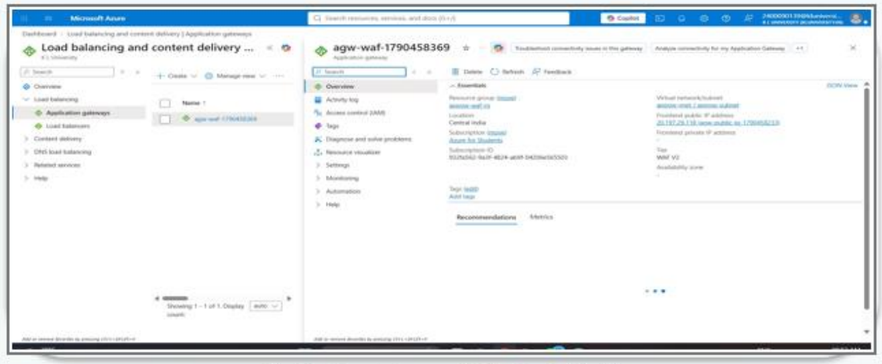
WAF SECURITY

Web Application Firewall (WAF) is a security layer that protects web applications from harmful and unwanted traffic. It checks incoming requests before they reach the backend servers. WAF can detect common web attacks such as SQL Injection and Cross-Site Scripting (XSS) and block suspicious requests.
It also allows normal users to access the application without interruption. In our project, WAF is first tested in Detection Mode to identify possible false positives. After testing and tuning, Prevention Mode can be used to block harmful requests and improve the security of the application.

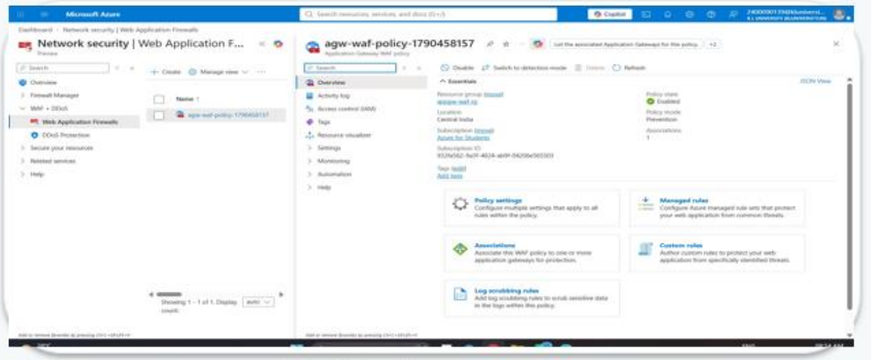
SYSTEM WORKFLOW

- **USER SENDS A REQUEST** - THE USER ACCESSES THE WEB APPLICATION THROUGH HTTPS.
- **APPLICATION GATEWAY RECEIVES THE REQUEST** - IT ACTS AS THE ENTRY POINT BETWEEN THE USER AND THE BACKEND SERVERS.
- **TLS TERMINATION** - THE GATEWAY HANDLES THE HTTPS CONNECTION SECURELY.
- **WAF CHECKS THE REQUEST** - WEB APPLICATION FIREWALL CHECKS THE INCOMING REQUEST FOR HARMFUL TRAFFIC SUCH AS SQL INJECTION OR XSS.
- **VALID REQUEST IS ALLOWED** - IF THE REQUEST IS SAFE, IT CONTINUES TO THE NEXT STEP. HARMFUL REQUESTS ARE BLOCKED.
- **URL-BASED ROUTING** - THE GATEWAY CHECKS THE URL AND SENDS THE REQUEST TO THE CORRECT BACKEND POOL.
- **BACKEND SERVER PROCESSES THE REQUEST** - THE SELECTED VM RECEIVES AND PROCESSES THE REQUEST.
- **HEALTH CHECK** - THE APPLICATION GATEWAY USES A HEALTH PROBE TO MAKE SURE THE BACKEND SERVER IS AVAILABLE.
- **RESPONSE IS SENT BACK** - THE BACKEND SERVER SENDS THE RESULT BACK THROUGH THE GATEWAY TO THE USER.

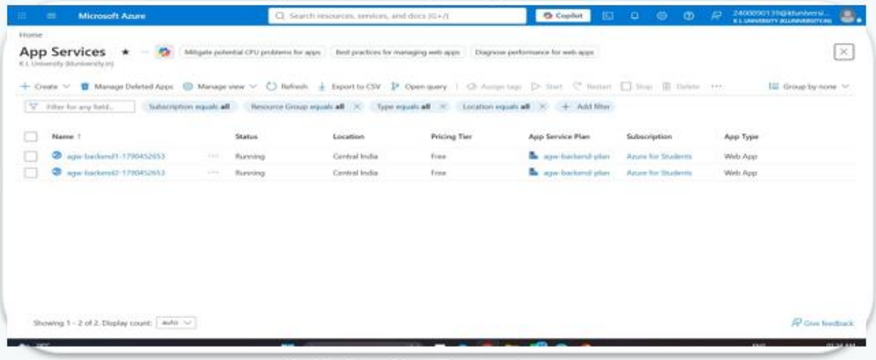
IMPLEMENTATION SCREENSHOTS



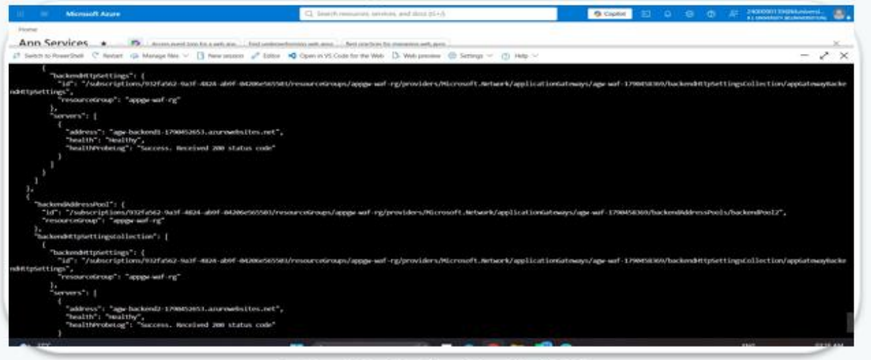
Application Gateway • WAF_v2 • Public IP



WAF Policy • Prevention • Enabled

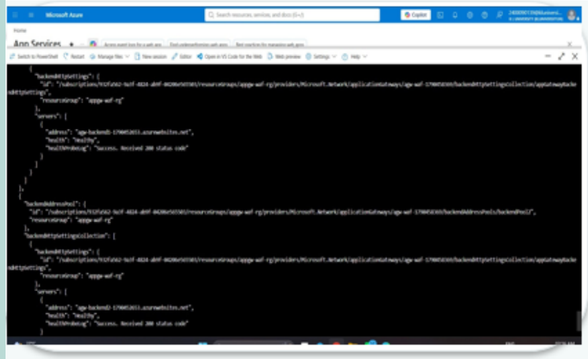


Two Backend App Services • Running



Backend Health • Healthy • HTTP 200

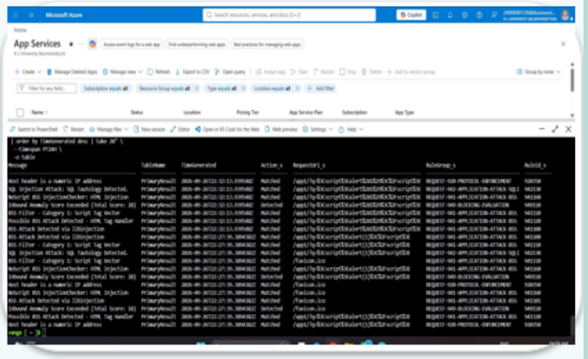
RESULT / OUTCOME



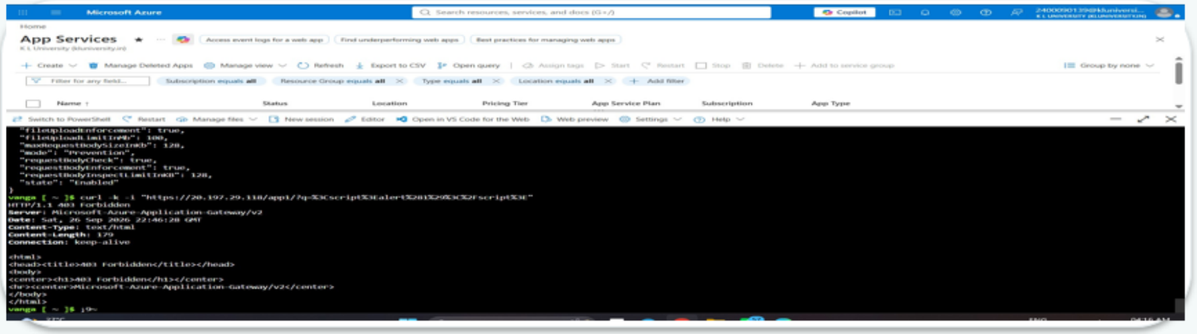
Live /app1/ → Backend Server 1



Live /app2/ → Backend Server 2



WAF Detection Logs • Rules Matched



WAF Prevention • HTTP 403 Forbidden

CONCLUSION

Azure Application Gateway with Web Application Firewall provides a secure and reliable way to manage web application traffic.

- WAF checks incoming requests and blocks harmful traffic.
- TLS termination helps secure HTTPS connections.
- URL path-based routing sends requests to the correct backend pool.
- Health probes check whether backend servers are available.
- The solution improves security, traffic management, and availability.

Future Scope

- Add more backend servers.
- Enable advanced monitoring and alerts.
- Deploy the application using Azure Kubernetes Service (AKS).